EXPERIMENT 7

main.py Python

```python
# ---------------- STATEFUL SYSTEM ----------------
# Example: Shopping cart that remembers previously added items

cart = []   # Global state (memory)

def stateful(item):
    cart.append(item)
    return f"Stateful Cart: {cart}"

n = int(input("Enter number of items to add to stateful cart: "))

for i in range(n):
    item = input("Enter item name: ")
    print(stateful(item))


# ---------------- STATELESS SYSTEM ----------------
# Example: System does not remember previous items

def stateless(item):
    cart = []      # Local variable (resets every time)
    cart.append(item)
    return f"Stateless Cart: {cart}"

n = int(input("\nEnter number of items for stateless cart: "))

for i in range(n):
    item = input("Enter item name: ")
    print(stateless(item))
```

Output:

PROBLEMS DEBUG CONSOLE TERMINAL PORTS OUTPUT zsh - expt_7

```
→  expt_7 git:(main) ✗ python3 main.py
Enter number of items to add to stateful cart: 3
Enter item name: jack
Stateful Cart: ['jack']
Enter item name: joe
Stateful Cart: ['jack', 'joe']
Enter item name: jill
Stateful Cart: ['jack', 'joe', 'jill']

Enter number of items for stateless cart: 3
Enter item name: jack
Stateless Cart: ['jack']
Enter item name: joe
Stateless Cart: ['joe']
Enter item name: jill
Stateless Cart: ['jill']
→  expt_7 git:(main) ✗
```